

torch.linalg.cond#

<https://pytorch.org/docs/stable/generated/torch.linalg.cond.html>

Description:

Computes the condition number of a matrix with respect to a matrix norm. Letting $K \in \mathbb{R}^n \times \mathbb{R}^n$ or $\mathbb{C}^n \times \mathbb{C}^n$, the condition number κ of a matrix $A \in \mathbb{K}^{n \times n}$ is defined as $\kappa(A) = \frac{\|A\|}{\|A^{-1}\|}$. The condition number of A measures the numerical stability of the linear system $AX = B$ with respect to a matrix norm. Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions. p defines the matrix norm that is computed. The following norms are supported:

Function Signature

```
torch.linalg.cond(A, p=None, *, out=None) → Tensor#
```

Parameters

Keyword Arguments

Returns

Raises

Parameters

Keyword

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

Returns:

A real-valued tensor, even when A is complex.

Code Examples

```
>>> A = torch.randn(3, 4, 4, dtype=torch.complex64)
>>> torch.linalg.cond(A)
>>> A = torch.tensor([[1., 0, -1], [0, 1, 0], [1, 0, 1]])
>>> torch.linalg.cond(A)
tensor([1.4142])
>>> torch.linalg.cond(A, 'fro')
tensor(3.1623)
>>> torch.linalg.cond(A, 'nuc')
tensor(9.2426)
>>> torch.linalg.cond(A, float('inf'))
tensor(2.)
>>> torch.linalg.cond(A, float('-inf'))
tensor(1.)
>>> torch.linalg.cond(A, 1)
tensor(2.)
>>> torch.linalg.cond(A, -1)
tensor(1.)
>>> torch.linalg.cond(A, 2)
tensor([1.4142])
>>> torch.linalg.cond(A, -2)
tensor([0.7071])

>>> A = torch.randn(2, 3, 3)
>>> torch.linalg.cond(A)
tensor([[9.5917],
        [3.2538]])
>>> A = torch.randn(2, 3, 3, dtype=torch.complex64)
>>> torch.linalg.cond(A)
tensor([[4.6245],
        [4.5671]])
```

Notes & Warnings

NOTE

Note When inputs are on a CUDA device, this function synchronizes that device with the CPU if `p` is one of ('fro', 'nuc', inf, -inf, 1, -1).

NOTE

See also `torch.linalg.solve()` for a function that solves linear systems of square matrices. `torch.linalg.lstsq()` for a function that solves linear systems of general matrices.

Detailed Documentation

Documentation

Computes the condition number of a matrix with respect to a matrix norm.

Letting $K \in \mathbb{R}^n \times \mathbb{R}^n$ or $\mathbb{C}^n \times \mathbb{C}^n$, the condition number κ of a matrix $A \in K^{n \times n}$ is defined as

The condition number of A measures the numerical stability of the linear system $AX = B$ with respect to a matrix norm.

Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions.

`p` defines the matrix norm that is computed. The following norms are supported:

2-norm (largest singular value)

`max(sum(abs(x), dim=1))`

`min(sum(abs(x), dim=1))`

`max(sum(abs(x), dim=0))`

`min(sum(abs(x), dim=0))`

largest singular value

smallest singular value

where `inf` refers to `float('inf')`, NumPy's `inf` object, or any equivalent object.

For `p` is one of ('fro', 'nuc', `inf`, `-inf`, 1, -1), this function uses `torch.linalg.norm()` and `torch.linalg.inv()`. As such, in this case, the matrix (or every matrix in the batch) A has to be square and invertible.

For `p` in (2, -2), this function can be computed in terms of the singular values $\sigma_1 \geq \dots \geq \sigma_n$

In these cases, it is computed using `torch.linalg.svdvals()`. For these norms, the matrix (or every matrix in the batch) A may have any shape.

When inputs are on a CUDA device, this function synchronizes that device with the CPU if `p` is one of ('fro', 'nuc', `inf`, `-inf`, 1, -1).

`torch.linalg.solve()` for a function that solves linear systems of square matrices.

`torch.linalg.lstsq()` for a function that solves linear systems of general matrices.

A (Tensor) – tensor of shape $(*, m, n)$ where $*$ is zero or more batch dimensions for p in $(2, -2)$, and of shape $(*, n, n)$ where every matrix is invertible for p in $(\text{'fro'}, \text{'nuc'}, \text{inf}, -\text{inf}, 1, -1)$.

p (int, inf, -inf, 'fro', 'nuc', optional) – the type of the matrix norm to use in the computations (see above). Default: None

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

A real-valued tensor, even when A is complex.

RuntimeError – if p is one of $(\text{'fro'}, \text{'nuc'}, \text{inf}, -\text{inf}, 1, -1)$ and the A matrix or any matrix in the batch A is not square or invertible.